Artificial Intelligence for Engineers

Lab Experiment 5: Regression Model to Predict Motor Temperature Rise

Name: Abhinav Kumar
Roll No: 1024031186
Course: AI for Engineers

Objective

To develop a Linear Regression model that predicts the temperature rise in an electric motor based on operational parameters such as voltage, current, ambient temperature, load, and running time.

Tools and Libraries Used

Python, NumPy, Pandas, Scikit■learn, Matplotlib, Jupyter Notebook

Dataset Description

A synthetic dataset containing 200 samples was generated using Python. The dataset represents different operating conditions of a motor and the corresponding temperature rise.

Feature	Description
Voltage	Input voltage supplied to the motor
Current	Electrical current flowing through the motor
AmbientTemp	Temperature of the surrounding environment
Load	Percentage load applied to the motor
RunningTime	Duration for which the motor runs
TempRise	Increase in motor temperature (target variable)

Methodology

1. The dataset was generated and stored using Pandas DataFrame. 2. Independent variables (Voltage, Current, AmbientTemp, Load, RunningTime) and the dependent variable (TempRise) were separated. 3. The dataset was split into training (80%) and testing (20%) sets. 4. A Linear Regression model from Scikit■learn was trained using the training data. 5. The model performance was evaluated using R^2 Score, MAE, and RMSE.

Model Evaluation Results

Metric	Value
R^2 Score	0.8437
Mean Absolute Error (MAE)	1.358
Root Mean Square Error (RMSE)	1.770

Prediction Example

For input conditions: Voltage = 230 V, Current = 10 A, Ambient Temperature = 30°C, Load = 80%, and Running Time = 60 minutes, the model predicted a temperature rise of approximately **49.92°C**.

Conclusion

The Linear Regression model successfully predicts motor temperature rise based on operational parameters. The model achieved an R^2 score of 0.84, indicating good predictive performance. The analysis shows that current and ambient temperature significantly influence motor heating. This model can help monitor motor performance and prevent overheating in practical engineering applications.